

jQuery Basics

What is jQuery?

- jQuery is a fast, small, and feature-rich JavaScript library
- It is used for
 - HTML document traversal and manipulation
 - event handling
 - animation
 - Ajax
- With a combination of versatility and extensibility, jQuery has changed the way that millions of people write JavaScript.

Why jQuery?

- jQuery is cross-browser
- jQuery is more easy to use than raw javascript
- jQuery is extensible
- jQuery is simplifies and has rich AJAX support
- jQuery has large development community and many plugins
- Excellent documentation

How to use jQuery in WebApp?

- Download jQuery from www.jquery.com
- Reference or link the same in your application just like external javascript file.

■

```
<!DOCTYPE html>
<html>
<head>
<script src="jquery/3.4.1/jquery.min.js"></script>
</head>
```

Javascript v/s jQuery

	JavaScript	jQuery
Type	Programming Language	API (Application Programming Interface)
Language	Written in C. Interpreted Language.	Uses resources given by JavaScript to make things easier.
Simplicity	One need to write entire code from scratch which is time consuming.	No need to write all code, scripts are already available.
Compatibility	Need to handle multi-browser compatibility.	It is cross-browser. No need to handle compatibility.
Length of Code	Need to write more code.	Code is less than javascript.
Light/Heavy Weight	Heavy weight compare to jQuery.	Lightweight

jQuery Syntax

jQuery syntax is made by using HTML elements selector and perform some action on the elements are manipulation in Dot sign(.).

```
$(document).ready(function() {  
    $(selector).action();  
});
```

- `$` sign define the jQuery.
- A (`selector`) defines the *Query element's* to find in HTML element's.
- And `action()` to be performed on the element's.

`$(document).ready(function())` in jQuery

- `$(document).ready` is a jQuery event
- It fires when DOM is fully loaded and ready to be manipulated by script
- Earliest process of page load process
- Script can safely access the elements of html DOM
- This events fires before all images, css etc. are fully loaded

Examples

```
<!DOCTYPE html>
<html>
<head>
  <title>jQuery hide p elements</title>
  <script src="jquery-latest.min.js"></script>
  <script>
    $(document).ready(function() {
      $("button").click(function(){
        $("p").hide();
      });
    });
  </script>
</head>
<body>
  <p>First paragraph start here...</p>
  <p>Second paragraph start here...</p>
  <button>Click here to hide above all paragraph</button>
</body>
</html>
```


jQuery Selectors

- jQuery selectors is most important aspects of the jQuery library.
- jQuery selector allow you to select and manipulate elements
- Selectors are useful and required at every step while using jQuery.

jQuery Selectors Syntax

Selector	Description
<u>element Selector</u>	Selects all element match of given elements.
<u>this Selector</u>	Selects current elements.
<u>#id Selector</u>	Selects element whose id is match of given elements.
<u>.class Selector</u>	Selects element whose class is match of given elements.
<u>*</u>	Selects all elements in the document.

Element Selector

- To select the element by html tag name we use Element Selector
- Syntax `$(element_name)`

`$( 'td' )` : Selects all td elements

`$( 'div a' )` : Select all anchor elements that are descendants of div

`$( 'div, span, a' )` : select all div, span and anchor elements

Element Selector

- To select **td** tag and count total number of **td** tags in document

```
<script type="text/javascript">
    $(document).ready(function () {
        alert($('td').length);
    });
</script>
```

- Select all **tr** tag and change its back-ground color

```
<script type="text/javascript">
    $(document).ready(function () {
        $('tr').css('background-color', 'red');
    });
</script>
```

This Selector

- To select the current HTML element we use *this* Selector
- Syntax \$('this')

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $(this).css("background-color", "yellow");
    });
});
</script>
```

#id Selector

- #id selector is most efficient selector among all jQuery selectors
- If you know the id of an element you want to select, always use #id selector
- HTML id is always unique on the page. jQuery #id selector returns only first element if you have more than one selector with name on single page.
- jQuery doesn't return any error when element with particular id not found unlike javascript.
- Syntax `$(#element id)`

#id Selector

This will change the background colour of button with id button1.

```
<script src="jquery-1.11.2.js"></script>
<script type="text/javascript">
    $(document).ready(function () {
        $('#button1').css('background-color', 'yellow');
    });
</script>
</head>
<body>
    <input id="button1" type="button" value="Click Me" />
</body>
</html>
```

.class Selector

- To select the element by their css class name class Selector is used
- Class Selector uses JavaScript *getElementByClassName()*
- Syntax `$('.class-name')`

`$('.small')` : Selects all elements with class small

`$('.small, .big')` : Select all elements with class small or big

`$('#div.small, .big')` : Select div elements with class small and any element with class big

.class Selector

- Select all elements with class small and set its border to 5px with red colour

```
<script type="text/javascript">  
    $(document).ready(function () {  
        $('.small').css('border', '5px solid red');  
    });  
</script>
```

[attribute] Selector

- To select the element by their attribute or attribute with specific value attribute selector is used
- Syntax `$( '[attribute]' )`
`$( '[attribute="value"]' )`

`$( '[title]' )` : Selects all elements having title attribute

`$( 'div [title]' )` : Select all elements having title attribute

`$( '[title]="tt1"' )` : Select all elements that have title attribute value - tt1

`$( 'div [title]="tt1"' )` : Select all div elements that have title attribute value - tt1

.class Selector

- Select all elements having title attribute and set its border to 5px with red colour

```
<script type="text/javascript">  
    $(document).ready(function () {  
        $('[title]').css('border', '5px solid red');  
    });  
</script>
```

* Selector

- * selector will select all the elements of DOM

Event Handling

- All the user action to which a web page can respond to are known as event
- Some common actions are
 - Click
 - Double click
 - Hover
 - Key press
- When event occurred a Event Handler is called

Binding an Event to Element

■ Using bind()

```
$(selector).bind(eventType,eventData,handlerFunction)
```

- ✓ eventType: JavaScript event like click, hover etc.
- ✓ eventData: Any data that you want to pass to event handler function when event fired(triggered)
- ✓ handlerFunction: the operation to be performed each time event is fired

Binding an Event to Element

- **Using event method directly()**

 - `$(selector).eventType(handlerFunction)`

 - ✓ `$(selector).click(handlerFunction)`

 - ✓ `$(selector).hover(handlerFunction)`

 - **Removing event**

 - ✓ `$(selector).unbind(eventType, handlerFunction)`

jQuery Effects

- The jQuery library provides several techniques for adding animation to a web page.
- These include simple, standard animations that are frequently used, and the ability to craft sophisticated custom effects.

jQuery Effects

<u>animate()</u>	Runs a custom animation on the selected elements
<u>fadeIn()</u>	Fades in the selected elements
<u>fadeOut()</u>	Fades out the selected elements
<u>fadeTo()</u>	Fades in/out the selected elements to a given opacity
<u>fadeToggle()</u>	Toggles between the fadeIn() and fadeOut() methods
<u>hide()</u>	Hides the selected elements

jQuery Effects

<u>show()</u>	Shows the selected elements
<u>slideDown()</u>	Slides-down (shows) the selected elements
<u>slideToggle()</u>	Toggles between the slideUp() and slideDown() methods

```
$(selector).animate({params},speed,callback);
```



Almost any CSS Property (color, background-color)



CSS Property should be in Camel Case

padding-left → paddingLeft

jQuery animate()

- The jQuery animate() allows us to animate CSS properties

`$(selector).animate(properties,[duration],[easing],[complete])`

- ✓ **Properties:** CSS Properties
- ✓ **Duration:** Duration of animation in milliseconds. Default is 400.
- ✓ **Easing:** Used for transition. Default is swing.
- ✓ **Complete:** A function to call once animation is complete.

jQuery callback()

jQuery jQuery Getters & Setter

- One very important part of jQuery is the possibility to manipulate the DOM.
- jQuery has DOM related methods that is used to to access and manipulate elements and attributes.
- A few of these methods are `text()`, `html()`, `attr()`, and `val()`

jQuery get/set content

- The jQuery **text()** method is either used to get the combined text contents of the selected elements, including their descendants, or set the text contents of the selected elements.

Syntax:

```
$(selector).text();           // for get content  
$(selector).text(new text);   //for set content
```

jQuery get/set content

Example

jQuery get/set content

- The jQuery `html()` method is used to get or set the HTML contents of the elements.

Syntax:

```
$(selector).html();           // for get html  
$(selector).html(html_code);  //for set html
```

jQuery get/set content

Example

jQuery get/set content

- The jQuery **val()** method is used to get or set the current value of the HTML form elements such as **<input>**, **<select>** and **<textarea>**.

Syntax:

```
$(selector).val();           // to get value  
$(selector).val(newVal);     //for set new value
```

jQuery get/set content

Example

jQuery get/set content

- jQuery **attr()** method to either get the value of an element's attribute or set one or more attributes for the selected element.

Syntax:

```
$(selector).attr();           // to get value
```

```
$(selector).attr({"attribute1":"attribute value1",  
"attribute2":"attribute value2"...});    //for set new value
```

jQuery add/remove CSS Class

- jQuery provides several methods, such as **`addClass()`**, **`removeClass()`**, **`toggleClass()`** etc. to manipulate the CSS classes assigned to HTML elements.

jQuery `addClass()`

- jQuery `addClass()` method adds one or more classes to the selected elements.

Syntax:

```
$(selector).addClass("class1 class2....");
```

jQuery removeClass()

- The jQuery **removeClass()** method used to remove the classes from the elements
- Using **removeClass()** method can remove a single class, multiple classes, or all classes at once from the selected elements.

Syntax:

```
$(selector).removeClass("class"); // remove single class
```

```
$(selector).removeClass("class1 class2"); // remove  
multiple class
```

```
$(selector).removeClass(); // remove all classes at once
```


jQuery toggleClass()

- The jQuery **toggleClass()** add or remove one or more classes from the selected elements
- If the selected element already has the class, then it is removed; if an element does not have the class, then it is added

Syntax:

```
$(selector).toggleClass("class1 class2..."); // Toggle  
one or more class
```

jQuery add/remove Elements

- jQuery provides several methods, like **append()**, **prepend()**, **before()**, **after()** etc. that allows us to insert new content inside an existing element.
- jQuery provides two methods, such as **empty()**, **remove()**, to remove existing HTML elements or contents from the document.

jQuery add Elements

■ **append()**

- jQuery append() method is used to insert content to the end of the selected elements.

■ **prepend()**

- The prepend() method is used to insert content to the beginning of the selected elements.

■ **before()**

- The jQuery before() method is used to insert content before the selected elements.

■ **after()**

- The jQuery after() method is used to insert content after the selected elements.

jQuery remove Elements

■ **empty()**

- jQuery empty() method removes all child elements as well as other descendant elements and the text content within the selected elements from the DOM.

■ **remove()**

- The jQuery remove() method removes the selected elements from the DOM as well as everything inside it.
- In addition to the elements themselves, all bound events and jQuery data associated with the elements are removed.

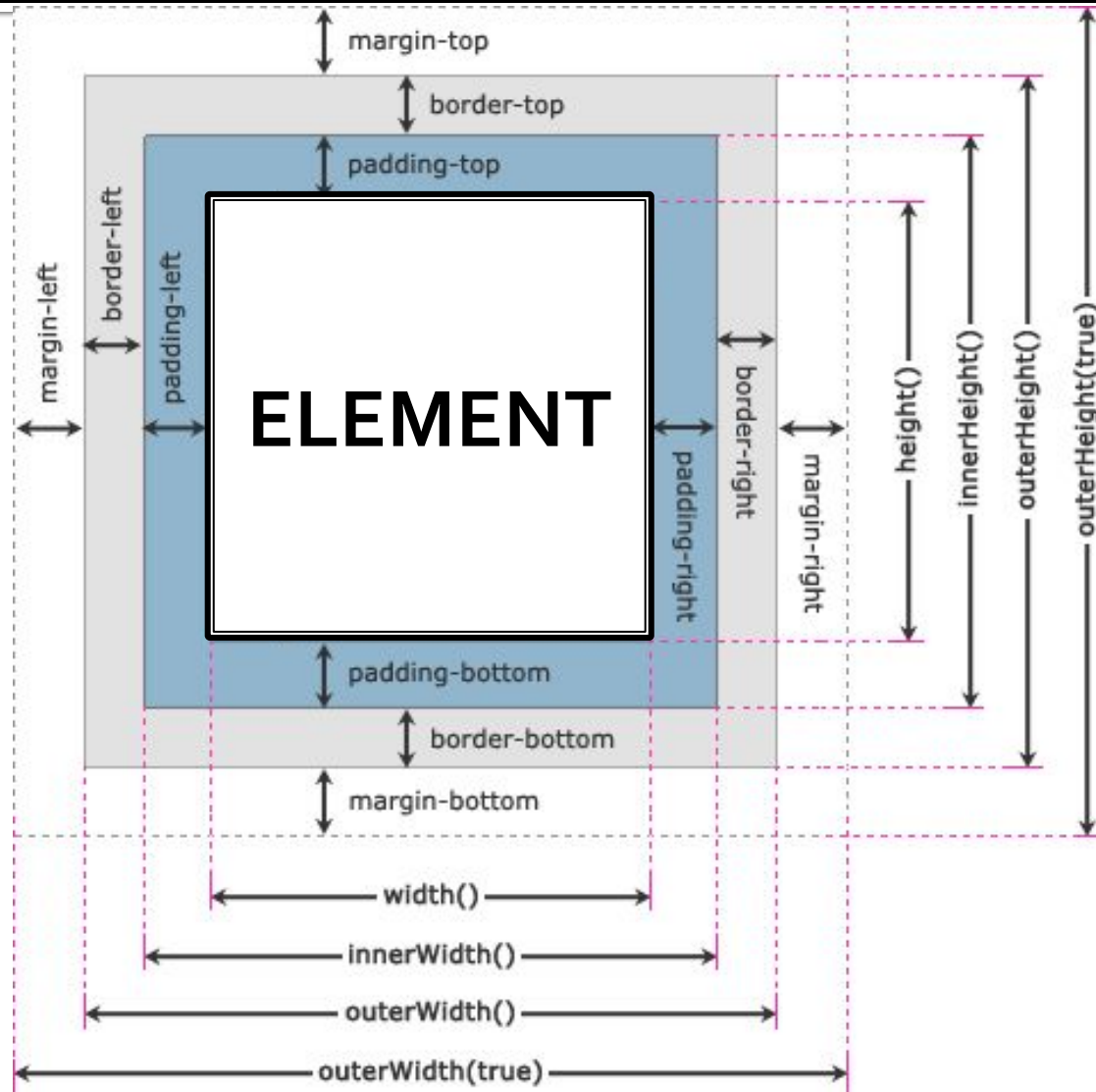
jQuery Dimensions

- jQuery provides several methods, such as

`height()`, `innerHeight()`, `outerHeight()`,
`width()`, `innerWidth()` and `outerWidth()`

to get and set the CSS dimensions for the elements.

jQuery Dimensions



jQuery Dimensions

height() & width()

- The jQuery width() and height() methods get or set the width and the height of the element respectively.
- This width and height doesn't include padding, border and margin on the element.

jQuery Dimensions

height() & width()

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        var divWidth = $("#box").width();
        var divHeight = $("#box").height();
        $("#result").html("Width: " + divWidth + ", " + "Height: " +
divHeight);
    });
});
</script>
```

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $("#box").width(400).height(300);
    });
});
</script>
```


jQuery Dimensions

innerHeight() & innerWidth()

- jQuery innerWidth() and innerHeight() methods get or set the *inner width* and the *inner height* of the element respectively.
- This inner width and height includes the padding but excludes *border* and *margin* on the element

jQuery Dimensions

innerHeight() & innerWidth()

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        var divWidth = $("#box").innerWidth();
        var divHeight = $("#box").innerHeight();
        $("#result").html("Inner Width: " + divWidth + ", " + "Inner Height: " + divHeight);
    });
});
</script>
```

jQuery Dimensions

innerHeight() & innerWidth()

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $("#box").innerWidth(400).innerHeight(300);
    });
});
</script>
```

jQuery Dimensions

outerHeight() & outerWidth()

- The jQuery `outerWidth()` and `outerHeight()` methods get or set the outer width and the outer height of the element respectively.
- This outer width and height includes padding and border but **excludes the margin** on the element.

jQuery Dimensions

outerHeight() & outerWidth()

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        var divWidth = $("#box").outerWidth();
        var divHeight = $("#box").outerHeight();
        $("#result").html("Outer Width: " + divWidth + ", " + "Outer Height: " + divHeight);
    });
});
</script>
```

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $("#box").outerWidth(400).outerHeight(300);
    });
});
</script>
```